



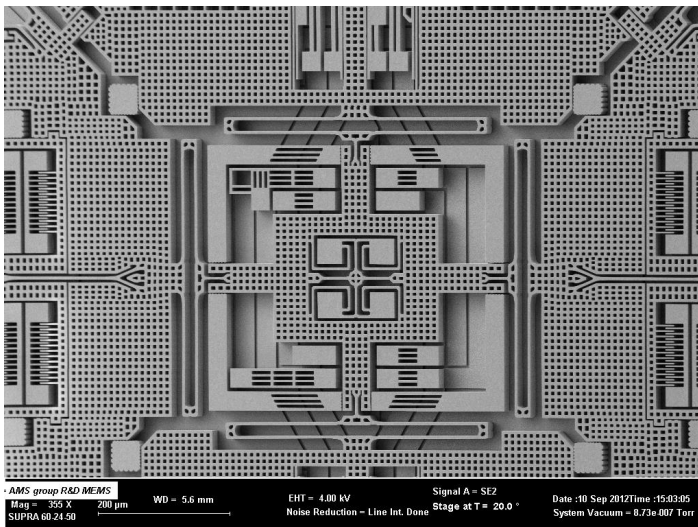
Sensors for AI-powered intelligent factories



Realizing intelligent factories through MEMS sensors

Intelligent factories help owners and operators to maximize the return on their investment in equipment, materials, capabilities, and human resources. A vision, an ideal, for many years, the intelligent factory is now possible through the availability of high-performance edge and data center computing power, ubiquitous sensing, and pervasive connectivity including wireless. These critical enabling factors are now augmented with advanced data science and artificial intelligence/machine learning (AI/ML), moving into the era of Industry 5.0.

Leveraging these developments, factories can maintain continuous awareness of critical processes and events, equipment status, materials, and the ambient environment. Gathering vast sensed-data channels into analytics applications, smart-factory systems can identify trends, make predictions, spot opportunities for improvement, and pre-empt and avoid issues such as equipment failures that can impair productivity in conventional enterprises.



Visit ST's MEMS technology webpage to find out more about our industry-leading ThELMA manufacturing process.

[MEMS technology webpage](#)

Today's state-of-the-art sensors permit the monitoring and measurement of numerous influential parameters. Among the most advanced are the tiny, low-power devices built with MEMS (Micro-Electro-Mechanical Systems) technology to capture accurate contextual and environmental data. These chip-size silicon devices are fabricated using wafer-level processes that ensure high accuracy, repeatability, and economies of scale in production. Enclosing tiny mechanisms and membranes sculpted in silicon, MEMS sensors provide functions such as inertial measurement, pressure sensing, temperature sensing, and analog and digital microphones. They can be installed in tiny modules, fitted to machines, inserted in tubes or chambers, integrated on circuit boards as part of embedded systems: anywhere precision sensing is needed.

Today's MEMS sensors offer high performance and accuracy at a fraction of the size and power budget of traditional sensors. System integrators can now deploy a wide variety of sensors, in massive numbers, to deliver capable and cost-effective intelligent-factory solutions for clients.

This whitepaper addresses the primary roles of some of these advanced MEMS sensors in digitalizing today's state-of-the-art intelligent factories.

The Industry 4.0 to Industry 5.0 transition

Industry 4.0, also known as the Fourth Industrial Revolution, is characterized by the integration of digital technologies into manufacturing processes. This includes the use of the Internet of Things (IoT), artificial intelligence (AI), big data analytics, and cyber-physical systems to create smart factories. The primary goals of Industry 4.0 are to enhance automation, improve efficiency, and enable real-time data exchange across the production chain.

However, as the digital landscape continues to evolve, a new paradigm is emerging, Industry 5.0. This next phase focuses on the collaboration between humans and machines, aiming to combine the efficiency of automation with the creativity and problem-solving capabilities of humans. The transition from Industry 4.0 to Industry 5.0 represents a significant shift in how we approach manufacturing and production. By focusing on human-machine collaboration, sustainability, and personalization, Industry 5.0 aims to create a more human-centric, resilient, and sustainable industrial landscape. This evolution not only enhances efficiency but also ensures that technology serves to augment human capabilities, leading to a more harmonious and productive future.

MEMS sensors play a pivotal role in Industry 5.0 by enabling precise and real-time monitoring of various industrial processes. These miniature sensors, which can measure parameters such as pressure, temperature, acceleration, humidity, and acoustic properties, facilitate the seamless integration of human and machine collaboration. For example, MEMS sensors enhance the capabilities of collaborative robots (cobots) by providing critical feedback for safe and efficient operation. Additionally, they support predictive maintenance and quality control by continuously collecting and analyzing data, thereby reducing downtime, and improving overall productivity. Their compact size, low power consumption, and high accuracy make MEMS sensors indispensable in creating a more intelligent, responsive, and human-centric industrial environment.

Industry 4.0			Industry 5.0		
Motion, vibration, angle measurement		Motion sensor inclinometers	Advanced interface		Human centric Helping individuals while working
Temperature monitoring & calibration		Temperature sensors	Artificial intelligence in the edge		
Pressure monitoring		Pressure sensor	Automation		Resilient Based on versatile technology
Humidity monitoring		Humidity sensors	Biosensors		Sustainable Conceived to optimize processes
Acoustic monitoring		MEMS microphones	Infrared sensor		

Figure 1: Comparison of Industry 4.0 and Industry 5.0

ST MEMS sensor properties: smart, open, accurate



ST's MEMS sensors' key attributes are that they are smart, open, and accurate; essential properties for advancing the integration of human and machine collaboration. These sensors should be designed with advanced intelligence to process data at the edge, enabling real-time decision-making and adaptive responses.

Their open architecture allows for seamless interoperability with other industrial sensors as a sensor hub and can configure itself according to the context needs, enabling a more connected and flexible manufacturing environment.

High accuracy ensures that the data collected is reliable, which is crucial for applications such as predictive maintenance, quality control, and safety monitoring. By combining these attributes, MEMS sensors contribute significantly to creating a more efficient, reactive, and human-centric industrial ecosystem.

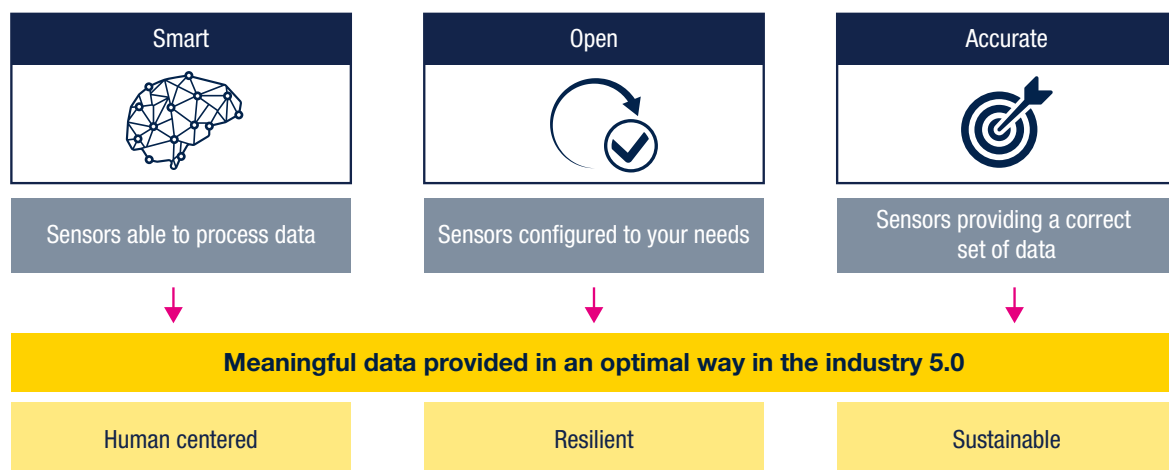


Figure 2. Key attributes of ST MEMS sensors for Industry 5.0

Intelligent factories enhance sustainability

Maximizing the efficient use of energy and resources such as water and materials consumed by manufacturing processes and day-to-day operation is at the heart of intelligent factory principles. ST's MEMS sensors provide data that can help track resources throughout the enterprise and the whole supply chain, measure consumption, pinpoint over-use, prevent production from generating scrap through poor process control, and minimize waste. Saving energy and resources, and promoting recycling, are crucial for achieving sustainability goals and fulfilling corporate social responsibility (CSR) obligations.

Intelligent factories, enabled by systems built with smart, accurate, open sensors, enhance long-term resource conservation and demonstrate commitment to ethical business practices, which is increasingly demanded by consumers, investors, and regulators. Opting for solutions that are efficient yet low in their own energy demands, such as ST's MEMS portfolio, maximizes the benefits to ensure investing in intelligent-factory technologies is both economically and environmentally sound.



Visit ST's Sustainability webpage to find out more about our commitments to sustainability, long-term goals and annual report.

[Sustainability webpage](#)

Industrial sensor longevity: ST's commitment to designers

ST backs-up all industrial components, including the sensors featured in this paper, with an ongoing, rolling commitment to continue production and support, typically up to 10 years into the future. This lets designers safely choose high-quality ST sensors, knowing that they will be available all along the application's lifecycle.



Visit ST's [longevity commitment program webpage](#) to find out more

Enabling the future factories of 2030 and beyond

A modern industrial plant is complex, and companies need to monitor many operations to ensure high productivity and safety. Real-time monitoring and sensing of the most important parameters at a plant is paramount to ensure successful execution and keep operational expenses to a minimum.

Automation, machine learning (ML), and AI are essential building blocks. They help the factory collect data through sensors by monitoring processes, products, and assets. Then, the factory's tools analyze the data first, to build a knowledge base that allows the factory and machinery to make timely decisions that can then help improve efficiency, reduce costs, and increase productivity.

A modern industrial system leveraging smart sensor nodes, gateways, and cloud infrastructure is designed to optimize efficiency and enable advanced analytics. Smart sensor nodes are intelligent devices equipped with MEMS sensors that collect and process data at the edge performing initial data filtering and analysis to reduce the volume of data transmitted.

Acting as intermediaries, gateways aggregate data from multiple smart sensor nodes and ensure secure and reliable communication with the cloud. They can perform additional data processing and protocol conversion, enabling seamless integration with different network types and industrial systems.

The cloud infrastructure provides scalable storage and powerful computing resources for advanced data analytics, machine learning, and long-term data archiving. It enables real-time monitoring, predictive maintenance, and remote management of industrial processes, offering immediate insights that drive informed decision-making and operational efficiency.

Together, these components create a robust and flexible system that enhances productivity, reduces downtime, and supports the seamless integration of human and machine collaboration in Industry 5.0. Figure 3 illustrates how data are collected, processed, and analyzed in a modern factory.

Enabling the future factories of 2030 and beyond, the characteristics of the latest ST MEMS sensors fully match the ongoing industrial evolution:

- **In-sensor** processing, leveraging machine learning, to analyze data quickly and efficiently.
- **Openness** that facilitates interfacing with other sensors, letting third-parties benefit from ST's on-sensor processing innovations while building an ecosystem to jointly create value for customers.
- **Accuracy** providing high quality data to drive informed decision making with smooth and natural interactions between people and machines. These sensors also reduce factory calibration time and resources, which also lowers energy needs and costs.

Smart sensors can continuously monitor equipment performance and detect anomalies or deviations from normal operation, allowing for early identification of potential issues before they escalate into major failures.

By preventing catastrophic failures and optimizing maintenance schedules, smart sensors can lead to significant cost savings in terms of repair costs, spare parts inventory, and overall operational efficiency.

The following sections describe five sensing modalities for equipment condition monitoring and general building management and planning in intelligent factories – vibration, inclination, sound, heat, and presence – and describe how to select and deploy ST's MEMS sensors.

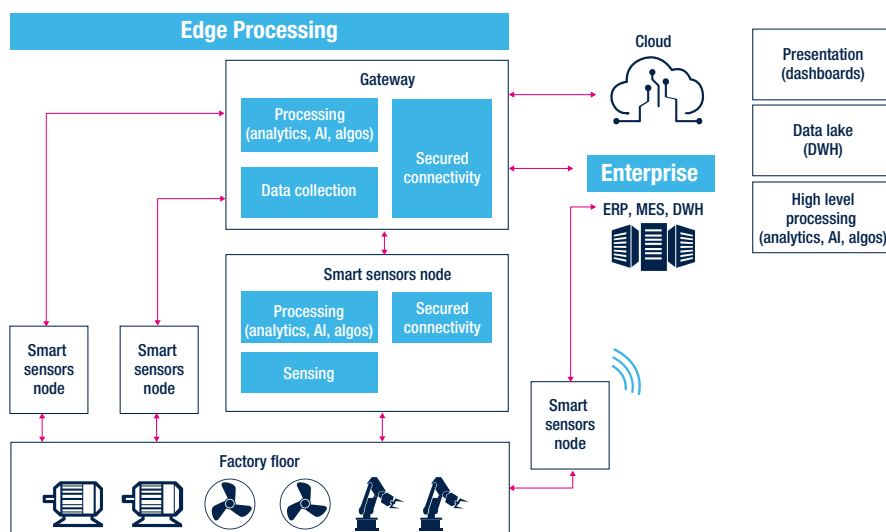


Figure 3. Sensors in a factory collect and analyze data to support decisions.

Vibration

Keeping equipment operational is crucial to maintaining optimal throughput and productivity of the plant. The onset of failure in industrial machinery is typically indicated by various symptoms. Considering a motor, the first of these is likely to be a change in the ultrasound signature. The next is vibration, and after a period, others such as audible noise and excessive heat. Each of these can be detected using MEMS devices such as microphones, accelerometers, and temperature sensors, which can detect signals of very small amplitude.

Vibration sensing is widely used to provide early failure detection in condition monitoring systems, enabling predictive maintenance and timely treatment of potential problems that can minimize line-down time.

Operators can deploy sensors to capture vibration data. Using software tools that embed known and proven analytical methods, such as Fast Fourier Transforms (FFT), they can quickly establish an operating baseline by running the machine for a short period to capture the frequency components of normal equipment vibration. Once defined, constant vibration sensing and analysis can detect abnormal patterns that can provide an early warning at the onset of any problems. These may be related to normal wear, to be addressed with routine adjustment or component replacement. On the other hand, the abnormal reading may indicate an issue elsewhere, such as a lubrication failure that can be fixed quickly and easily with minimal machine downtime. Figure 4 shows the pyramid of steps involved in vibration monitoring.

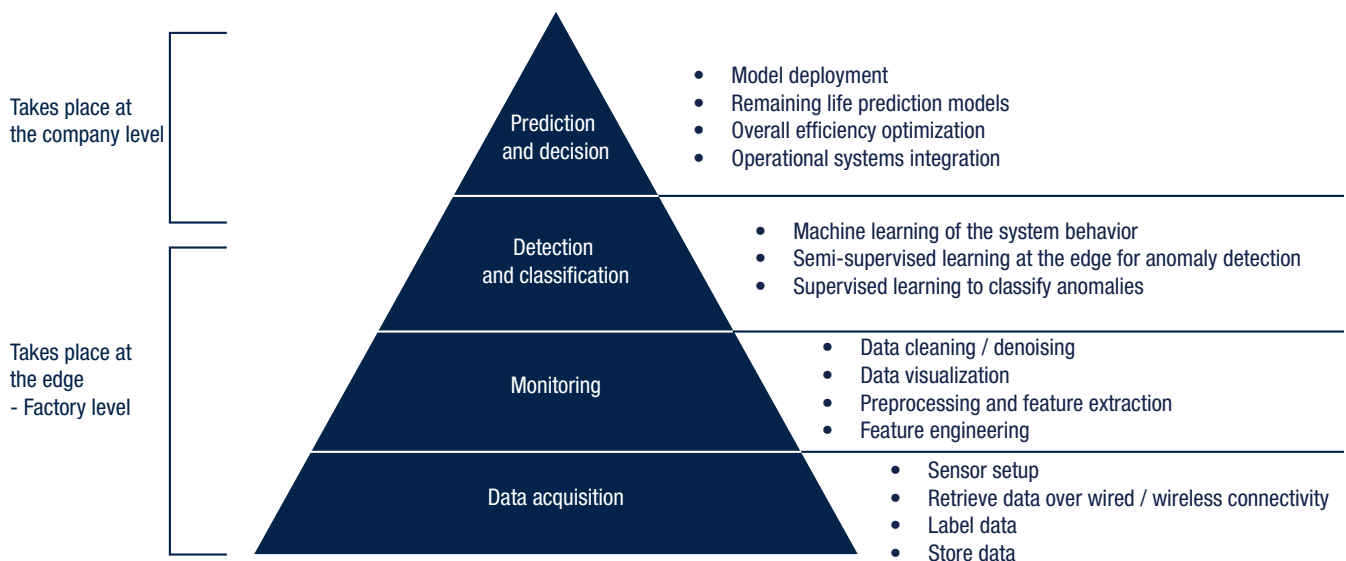


Figure 4. A pyramid of vibration monitoring highlights the various stages of machine-learning model development throughout the factory.

The latest innovative MEMS sensors, equipped with in-sensor processing, can sense and also analyze many parameters that help improve system energy efficiency, reduce cost, and increase productivity. Smart, open, and accurate sensors are cost-effective and consume very little power, making them ideal for use in intelligent factories.

Industrial automation retrofit with an accelerometer

Our latest innovation, the **IIS2DULPX ultralow-power accelerometer**, is set to transform the landscape of industrial applications. A robotic arm, for example, can now be retrofitted with a battery-operated sensor node designed to monitor vibration and shocks, which can significantly impact the integrity and reliability of the arm's tasks.

Thanks to its edge AI processing capabilities with a machine learning core (MLC) and adaptive self-configuration, the IIS2DULPX ensures accurate and continuous monitoring without the need for frequent battery recharging or replacement. Specifically tailored for battery-operated industrial IoT sensor nodes, its ultra-low power consumption and on-board intelligence ensures long-lasting operation (up to several years, in some cases), significantly reducing maintenance needs and enabling retrofitting of equipment that requires infrequent maintenance. Featuring self-configurability functions, the sensor is able to optimize its settings in real-time without the need for a host processor. This real-time adaptability ensures power savings without compromising the quality of sensing.

Ideal for harsh environments, the IIS2DULPX boasts an extended temperature range, operating reliably up to 105°C. It is the only accelerometer capable of offering ultra-low power and edge processing in such challenging conditions, making it an enabler for the new generation of industrial applications.

WHY ST MEMS SENSORS FOR VIBRATION MONITORING

The bandwidth of a MEMS sensor is crucial for vibration monitoring because it determines the range of frequencies that the sensor can accurately detect and measure. In industrial applications, machinery and equipment often operate at various speeds and under different conditions, producing vibrations across a wide spectrum of frequencies. ST has developed sensors that can capture vibration frequencies up to several kilohertz [1] according to the use case.

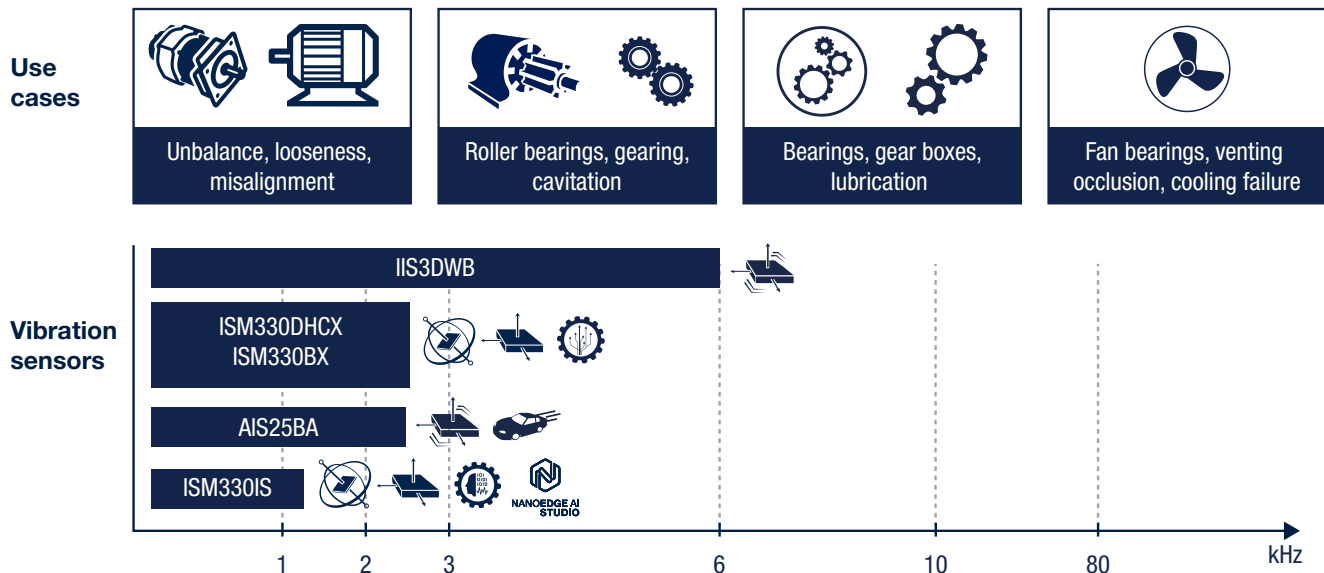


Figure 5: ST MEMS sensors for vibration and condition monitoring.

ST's IMUs such as the **ISM330DHCX**, **ISM330BX**, **ISM330IS(N)**, and accelerometers like the **IIS3DWB**, are particularly effective for industrial applications, with their extended temperature range and minimal power consumption. All these sensors (Table 1) are engineered to deliver high performance within the stated bandwidth, making advanced sensing capabilities available for a wide range of industrial uses.

The extended temperature range offered in the IIS3DWB accelerometers and ISM330DHCX IMU ensures reliable operation for vibration monitoring even in harsh and fluctuating environmental conditions. This is critical for industrial settings where equipment often faces extreme temperatures.

On the other hand, the IMUs combine ultra-low power consumption with in-sensor processing. Their outstanding energy efficiency can deliver cost savings and extend operational lifespans for battery-powered IIoT devices.

	Bandwidth [kHz]	Temperature range [°C]	In-sensor AI	Sensor fusion low power	Package size [mm]
IIS3DWB	6	-40 to +105	-	-	2.5x3x0.86
ISM330BX	2	-40 to +85	MLC and FSM	Yes	2.5x3x0.71
ISM330DHCX	2	-40 to +105	MLC and FSM	-	2.5x3x0.86
ISM330IS	1	-40 to +85	ISPU	Yes (with library)	2.5x3x0.86

Table 1: ST industrial products for vibration monitoring

IMPLEMENTING EDGE AI: ANOMALY DETECTION EXAMPLE

Anomaly detection and vibration monitoring are integral components of a robust predictive maintenance strategy. Vibration monitoring provides the essential data needed for anomaly detection, while anomaly detection techniques analyze this data to identify potential issues early. Anomaly detection involves detecting abnormal behavior or patterns in collected data. This can be performed inside the sensor; ST MEMS sensors contain dedicated technologies such as the machine-learning core (MLC) or intelligent sensor processing unit (ISPU). Using tools available through ST's MEMS Studio design software, developers can program the in-sensor MLC or ISPU to detect anomalies, working through the sequence described:

1. Collect data: The first step is to collect data from sensors.
2. Label data: The collected data may need to be preprocessed to remove noise, filter out irrelevant data, or transform the data into a suitable format for analysis.
3. Train a model: Anomaly detection models can be trained using machine learning algorithms such as clustering, classification, or regression. These models can learn patterns in the data and identify anomalies based on deviations from those patterns.
4. Deploy the model: once the model is trained, it can be deployed to the sensor to perform real-time anomaly detection.
5. Monitor the system: The system should be monitored regularly to ensure that the model is performing as expected. If the model detects an anomaly, it can trigger an alert or take corrective action, such as shutting down the machine or sending a notification to a technician.

Overall, anomaly detection in an edge solution requires collecting and preprocessing data, training a model, deploying the model to the edge device, and monitoring the system for anomalies.

By providing a choice of devices equipped with an embedded MLC or ISPU, ST gives developers flexibility to find the most effective solution in terms of computing capacity and flexibility in programming.

In either case, the in-sensor AI of ST's advanced MEMS sensors offers significant advantages for anomaly detection by enabling real-time, on-device analysis of data, which enhances the efficiency and responsiveness of industrial systems. Processing data directly within the sensor offloads the host CPU, extending energy-saving operation and minimizing latency thereby ensuring immediate detection and response to anomalies. Further advantages include reduced reliance on cloud-based processing, which can be slower and less secure, as well as conserving bandwidth by transmitting only relevant data, thereby optimizing network resources. The ability to perform advanced analytics at the edge also enhances the robustness and reliability of the system, ensuring continuous monitoring and swift identification of potential issues.



Inclination

The vertical and horizontal orientation of a machine is often an important specification in its use and can directly impact its performance. Unintended deviation from the ideal inclination, which can be caused by events such as vibrations or jostling, can cause problems that might lead to production shutdowns.

Recent advancements in MEMS technology now enable tiny high-performance, high-precision sensors that can collect data for dedicated algorithms to analyze for real-time position monitoring [2].

The inclination of an equipment or a robotic arm can be measured using either a static or dynamic measurement approach. A static inclinometer measures, using the projection of the earth gravity, the angle of inclination of a structure or object. Using a static inclinometer, the measure is reliable as long as there is no interfering acceleration, for example associated with sudden movement. A dynamic inclinometer measures reliably the angle of inclination of a structure or object even in presence of sudden movement, thanks to the fusion of information from sensors working on different principles (gyroscope and accelerometer). Static inclinometers are typically used for monitoring the stability of structures such as buildings, bridges, and dams, while dynamic inclinometers are used for monitoring the inclination in objects which move.

ST's **IIS2ICLX** two-axis linear accelerometer with digital output is the industry's first inclinometer with an embedded machine-learning core, bringing greater accuracy and greater energy efficiency within reach at the same time. It can execute machine learning applications based on decision trees with minimal power consumption. On the other hand, the sensing element offers ultra-low noise ($15 \mu\text{g}/\sqrt{\text{Hz}}$) and an offset change of less than $0.075 \text{ mg}/^\circ\text{C}$ temperature change. With its excellent signal-to-noise ratio, it is widely used to measure static or quasi-static angles, as well as low-frequency vibrations. The ceramic package and extended operating temperature range (-40 to $+105^\circ\text{C}$) help pass the most stringent industrial requirements.

The **IIS3DHHX** and **ISM330DHCX** are advanced sensors that can be used as static and dynamic inclination measurements

in various industrial applications. The **IIS3DHHX** is a high-resolution, low-noise accelerometer that excels in detecting subtle changes in orientation, making it ideal for applications requiring accurate tilt and motion sensing. On the other hand, the **ISM330DHCX** is a versatile IMU providing comprehensive motion tracking and orientation data with in-sensor AI capabilities. Together, these sensors offer robust and reliable solutions for monitoring and controlling the position of machinery, equipment, and structures, ensuring optimal performance and safety in industrial environments. Their high accuracy, low power consumption, and extended temperature range make them suitable for a wide range of applications such as antenna positioning, industrial robots and cobots, equipment installation, leveling instruments, and structural health monitoring systems.



Figure 6. Dynamic inclinometers are effective in monitoring the position and movement of an industrial robot.

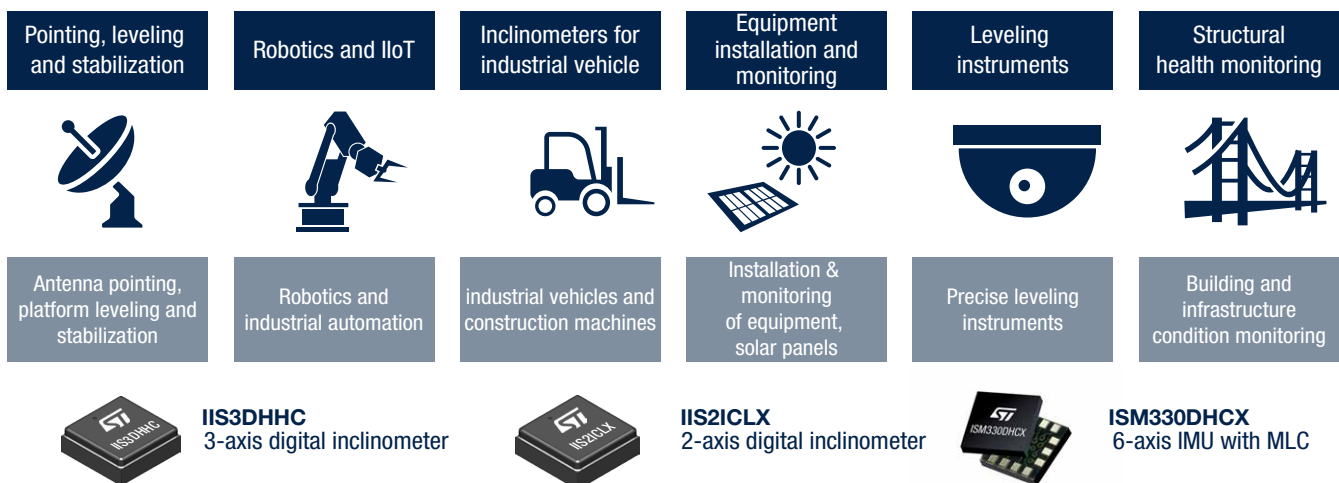


Figure 7. The MEMS inclinometer portfolio

Sound



The sound a machine generates during operation can be a major characteristic of declining performance. Moreover, as the frequency of that sound can extend into the ultrasonic range, it isn't always audible to humans. Sound anomaly, detected by appropriate sensors, is an important indicator for operators to monitor.

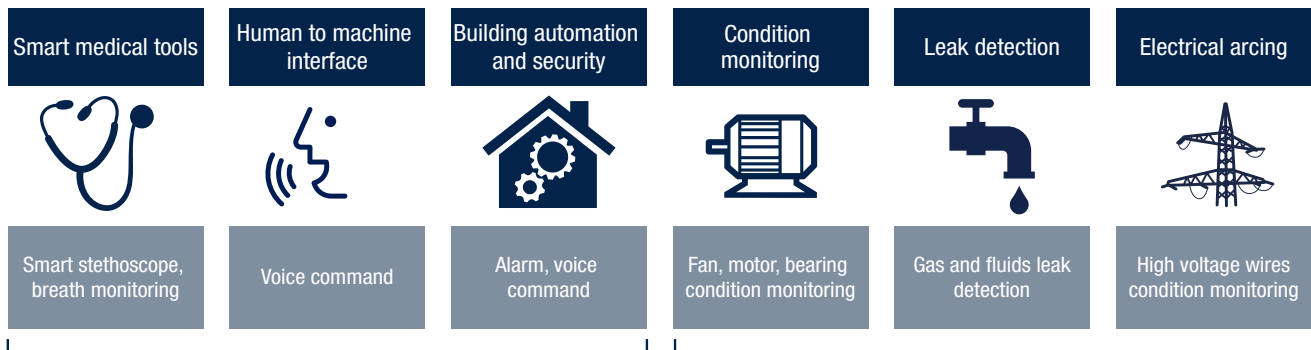
Early-stage leak detection in pipes delivering chemical gases, for example, is critical to avoiding catastrophic damage. Because the sound from a leak usually starts in the ultrasonic range, above our ability to hear, a factory should choose a dedicated sensor to sense these ultrasonic acoustic waves.

By analyzing the sound patterns of machinery or equipment over time, maintenance professionals can identify changes in sound that may indicate a problem. This can be done using specialized software and sensors that can detect and analyze sound frequencies and patterns. For example, if a machine is producing a high-pitched whine that was not present before, it could indicate that a bearing is starting to wear out. By detecting this early, maintenance professionals can schedule a repair or replacement before the bearing fails completely,

avoiding costly downtime and potential safety risks.

Overall, sound analysis can be a valuable tool in predicting maintenance needs and preventing equipment failure.

ST's latest MEMS microphones offer performance and cost advantages for sensitive ultrasonic sound detection. These microphones, **IMP34DT05** and **IMP23ABSU**, can also support voice-command applications [3]. Key advantages of MEMS microphones are easy system integration, small size, high signal-to-noise ratios, and affordability. MEMS microphones are well suited to applications aiming to detect compressed fluid leaks, vacuum leaks, steam trap failures, bearing condition monitoring, electrical arcing/tracking, fan, and motor unbalance. Noise in the ultrasound range is often the first signal that indicates that the machine's condition is starting to change. Operators can also use sensing and monitoring solutions equipped with MEMS microphones and speech recognition algorithms to operate and control equipment via voice command.



IMP34DT05
Top port digital microphone



IMP23ABSU
Analog bottom port single ended microphone

Figure 8. ST MEMS sensor portfolio for noise analysis for smart factories.

Heat



All factory equipment is designed to operate within a predefined temperature range. Friction from failing mechanical components in a machine typically generates undesired heat that can further damage equipment parts. At this point, the equipment is very close to final failure. In a worst-case scenario, this heating could lead to sparks and fire.

Infrared cameras and temperature sensors are useful to create thermal profiles. ST has accurate and affordable temperature sensors such as the [STTS22H](#), which can be installed on the equipment to monitor the operating temperature range [4] in real time. Then, sudden increases in the equipment's temperature would lead the sensor to generate an interrupt signal to trigger measures to avoid equipment failure.

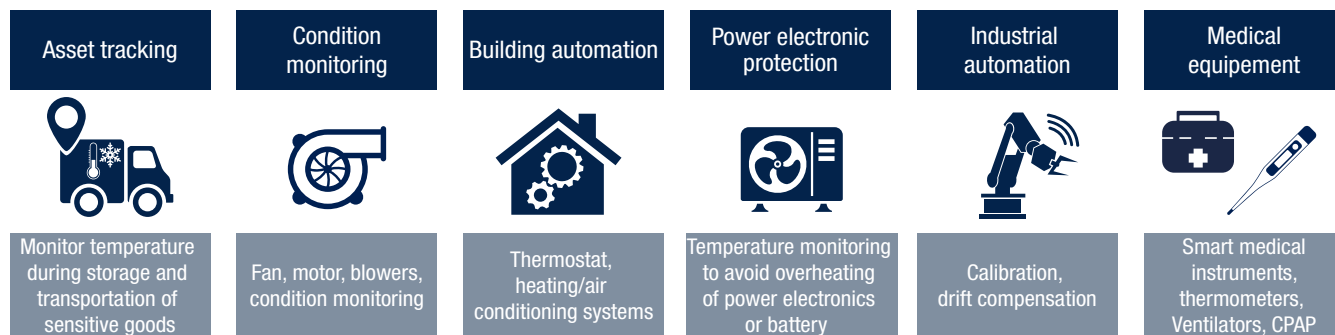


Figure 9. Market verticals where the heating measurement is a key factor.

Sensing presence

Detecting the presence of people can provide key information for enterprise resource planning. Sensors can efficiently detect the number and location of people in a factory to optimize energy consumption, as well as increase safety and security.

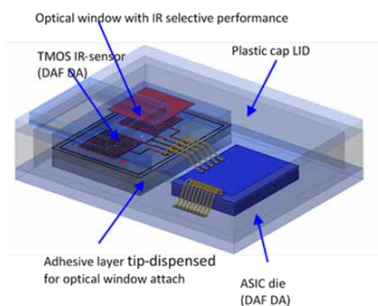
A presence detector consists of hardware and software tools and can be an expensive investment in a modern factory. The number and type of presence detection sensors will ultimately determine the features in the solution. Knowing the use cases for presence detection helps companies justify the development cost as part of an overall enterprise resource-planning solution.



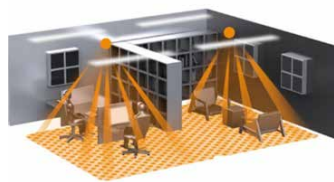
Intelligent HVAC (Heating, Ventilation, and Air Conditioning) control and automated lighting shutoffs use presence detection to optimize energy and enable significant savings. There are already proposed solutions to reduce energy consumption, including using multiple thermal imagers [5].

Thermal imaging using infrared detection is an innovative solution for presence detection [6]. Figure 10a shows the high-level design structure of ST's solution. Sensors able to operate in the 5 to 20 μm wavelength range of the electromagnetic spectrum can detect human presence by accurately sensing the amount of infrared radiation emitted from an object in the field of view. The information collected by the sensor can then be digitally processed by dedicated circuits embedded inside the tiny chip. The device offers the programmability to monitor motion, presence, or over-temperature conditions. Without a lens, it can detect human presence up to 4 meters away.

A successful presence detector fuses algorithms to process the sensor data with artificial intelligence algorithms to meet the demands of a 21st-century factory and buildings.



a) Sensor architecture



b) Sensor for presence and occupancy detection



c) Sensor for thermal profile as part of condition monitoring

Figure 10. High-level structure showing innovative IR sensors for occupancy and presence detection.

ST's **STHS34PF80** is an uncooled infrared sensor for occupancy detection applications. Unlike traditional PIR sensors, the STHS34PF80 detects both stationary and moving objects within its 80° field of view. It achieves this feature thanks to its high sensitivity (2000 LSB/°C) in the 5 to 20 μm wavelength range.

This sensor can detect human presence up to 4 meters away without requiring any additional external Fresnel lenses and simplifies integration thanks to a tiny LGA package (3.2 x 4.2 x 1.455 mm). Embedded algorithms within the sensor itself handle presence and motion detection, reducing reliance on external microcontrollers. This contributes to the sensor's ultra-low power consumption of just 10 μA at 1Hz, promoting sustainable product design.



Developer tools for ST industrial MEMS sensors

ST offers a complete ecosystem to support development and fast prototyping for industrial applications. From hardware tools to application examples, every developer can find the right solution and reduce time-to-market and related costs



- **Evaluation:** by combining the ready-to-use MEMS professional development platform ([STEVAL-MKI109D](#) and [STEVAL-MKI109V3](#)) and the adapter DIL 24 (listed in Table 2) you can quickly and easily evaluate the IMU leveraging the [complete MEMS-Studio desktop software development solution](#) (MEMS-Studio) that is downloadable for free. Play with the sensor, collect data, and perform all the analyses you are curious about.
- **Development:** the [STM32 Nucleo industrial expansion board for MEMS motion and audio sensors](#) (X-NUCLEO-IKS02A1) was developed relying on the [STM32 Open Development Environment](#) (STM32 ODE), an open, flexible, easy, and affordable way to develop innovative applications based on STM32 32-bit microcontrollers. It enables fast prototyping with leading-edge components that can be quickly transformed into final designs.

Using the MEMS Studio software and the [sensor and motion algorithm expansion software for STM32Cube](#) (X-CUBE-MEMS1), a software package that contains a collection of libraries ready to be used (including calibration and many other libraries), and some examples of source code for your development. With the X-CUBE-MEMS1 you can reduce your development effort by using the available examples and customize them for your specific application.

- **Prototype:** The [STWIN.box](#) development kit and reference design (STEVAL-STWINBX1) simplifies prototyping and testing of advanced industrial sensing applications in IoT contexts such as condition monitoring and predictive maintenance. The STWIN.box features a higher mechanical accuracy in the measurement of vibrations, an improved robustness, an updated BoM to reflect the latest and best-in-class MCU and industrial sensors, and an easy-to-use interface for external add-ons. The STWIN.box kit consists of an STWIN.box core system, a 480mAh LiPo battery, an adapter for the ST-LINK debugger, a plastic case, an adapter board for DIL 24 sensors, and a flexible cable. The many on-board industrial-grade sensors and the ultra-low power MCU enables applications that feature ultra-low power, 9 DoF motion sensing, wide-bandwidth vibration analysis, audio and ultrasound acoustic inspection, very precise local temperature, and environmental monitoring. A rich set of software packages is available in source code. Optimized firmware libraries and a complete companion cloud application help to speed up the design cycle to develop end-to-end solutions. The kit supports a broad range of connectivity options, including the built-in RS-485 transceiver, BLE, Wi-Fi, and NFC. The STWIN.box is suitable for field trials, demonstrations, and PoC for industrial IoT applications that use ST software and third-party software.

Product name	Evaluation board
IIS2DULPX	STEVAL-MKI246KA
IIS3DWB	STEVAL-MKI208V1K
ISM330DHCX	STEVAL-MKI210V2K
ISM330BX	STEVAL-MKI245KA
ISM330IS	STEVAL-MKI230KA
ISM330ISN	STEVAL-MKI233KA
IIS2ICLX	STEVAL-MKI209V1K
IIS3DHHX	STEVAL-MKI216V1K
IMP34DT05*	STEVAL-MIC003V1
IMP23ABSU*	STEVAL-MIC007V1
STTS22H	STEVAL-MKI200V1K
STHS34PF80	STEVAL-MKI231KA

Table 2: Industrial adapter DIL24 list

* Not supported by STEVAL-MKI109V3

In our MEMS sensors [GitHub repository](#), you will find sensor device drivers and application examples both for MLC and FSM, such as tilt detection, vibration monitoring, fall down, human activity recognition, head gestures, and more. To get started quickly with each example, the README file provides detailed information.

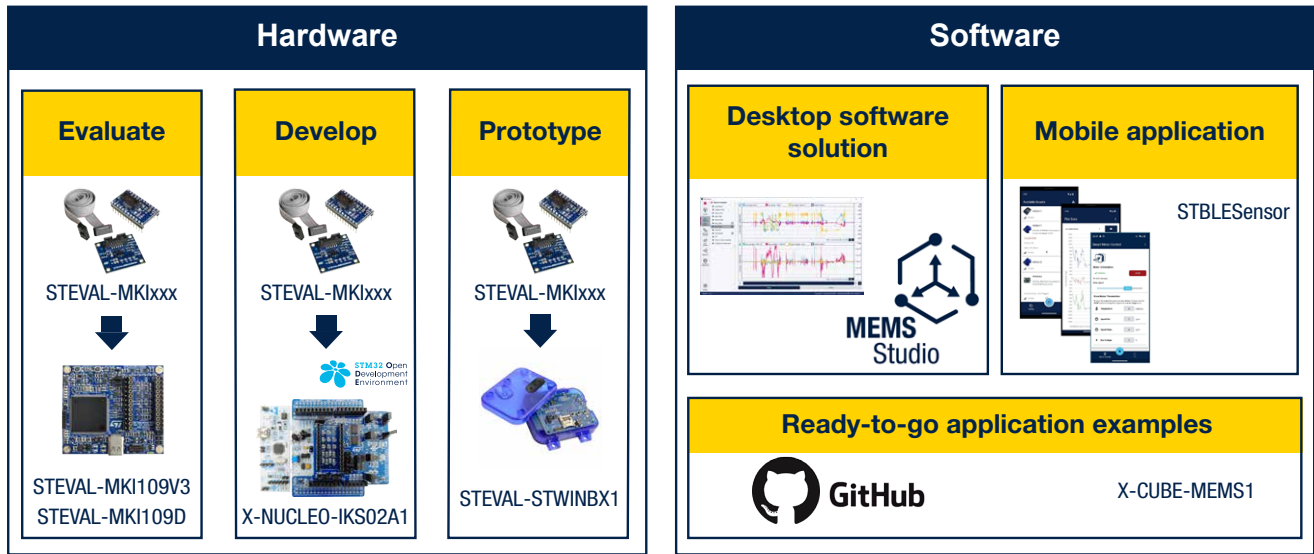


Figure 10: Shorten time-to-market with the ST ecosystem for industrial applications.

For the two IMU products with integrated ISPU, [ISM330IS](#) and [ISM330ISN](#), the ecosystem has additional specific tools to easily program the DSP. Embedded developers without any data science skills can use the [NanoEdge AI Studio](#) to program the ISPU (ISM330ISN), you can readily obtain accurate intelligence solutions with a limited amount of time and effort. For expert developers, the [ISPU Toolchain](#) helps to translate the C code in a register configuration for the ISPU.

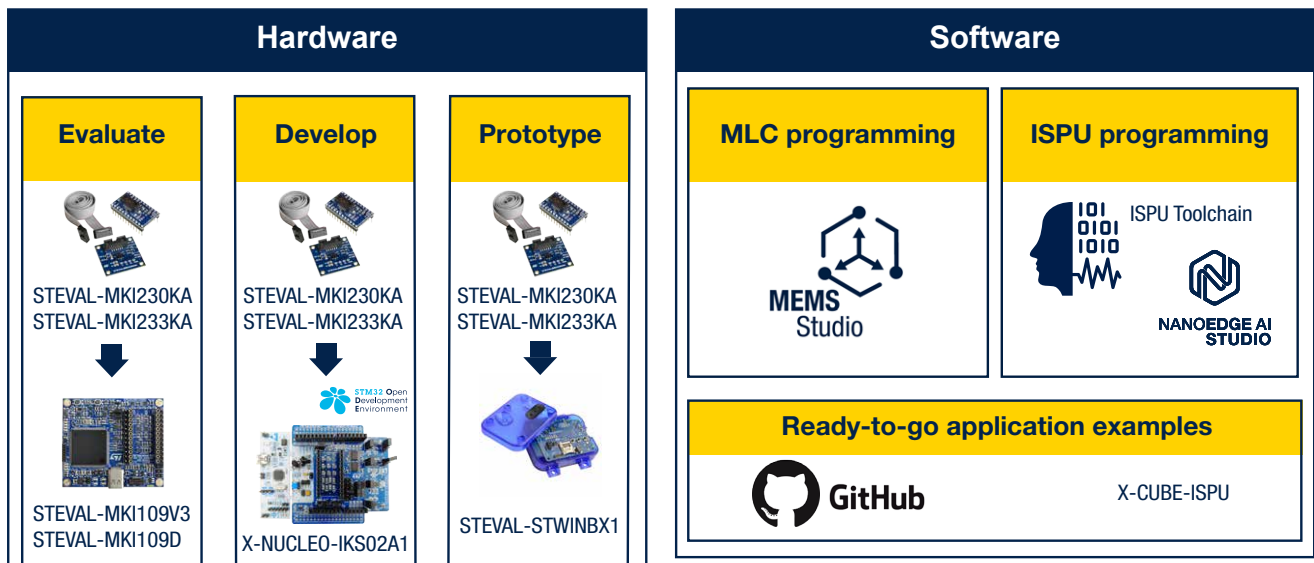


Figure 11: Shorten time-to-market with the ST ecosystem for industrial MEMS sensors with ISPU integrated.

Future trends



Deploying ST MEMS smart sensors with Edge AI in an industrial plant allows continuous collection of real-time data to monitor the operational status and condition of products and equipment.

In the most efficient factories, companies are using sensors to optimize efficiency, increase productivity, improve safety, and advance security. Still, in addition to sensors, companies need to continue adding comprehensive software tools for data analytics. Dedicated solutions will process and analyze sensor data using artificial intelligence to build models that can make operational and production decisions. Technologies and solutions for edge and cloud computing will play a critical role in implementing enterprise resource planning and execution.

ST MEMS sensors play a pivotal role in advancing Industry 5.0 by seamlessly integrating cutting-edge sensing technology with intelligent data processing. These sensors, including advanced IMUs, accelerometers, microphones, temperature, IR sensors, and more, offer high accuracy, low power consumption, and extended temperature ranges, making them ideal for a wide array of industrial applications.

By enabling real-time monitoring, predictive maintenance, and enhanced human-machine collaboration, ST MEMS sensors contribute significantly to creating more efficient, responsive, and sustainable industrial environments. Their ability to provide precise vibration, motion and position detection, combined with in-sensor AI capabilities, and their open architecture ensures that industrial systems are not only smarter but also more resilient and adaptable.

As Industry 5.0 continues to evolve, ST MEMS sensors will remain at the forefront, driving innovation and supporting the development of a more connected and human-centric industrial landscape.

Resources

ADDITIONAL DEVELOPER RESOURCES

ST MEMS and sensors [\[Product portfolio page\]](#)

Artificial intelligence at the edge [\[Technology page\]](#)

Micro-electro-mechanical systems (MEMS) process technology [\[Technology page\]](#)

MEMS sensors ecosystem for machine learning [\[Ecosystem page\]](#)

ST edge AI suite: your stepping stone to enabling edge AI [\[Ecosystem page\]](#)

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